

Introduction to Computer Architecture

Assignment 3

Due June 30, 2026

Grading Policy:

The ultimate goal for all presentation and assignments is review and understand. Assignment 3 aims for a comprehensive coverage of textbook materials for exam preparation.

Only (self-chosen) five questions of Assignment 3 are required ($100 = 20 \times 5$). Please specify the five questions you would like to use for grading.

You are highly encouraged to understand and answer the remaining questions toward a comprehensive review for the final exam.

Embrace now.

Good luck.

CHAPTER 1 - FUNDAMENTAL

01. What are the two critical performance techniques featured by the RISC architecture? Please explain also how they help to improve performance.
02. What are the key observations by the Denard Scaling Law and the Moore's law. Please explain also why they cease applying to more advanced CPU designs.
03. What is the conclusion of the Amdahl's Law. Please provide examples of using the Amdahl's Law to compute speedup for a program with one or more portions of parallelism-enhanced code.
04. What are the four classes of parallel architectures? Please provide also example design techniques if any for each class.
05. Please provide examples of memory addressing modes to showcase the usage of registers, immediates, and displacements.
06. What are the three impacts for computer cost to cease even without technology improvement?
07. Please provide an example to showcase the differences among faults, failures, and errors in a system.
08. How to computer MTTF of the entire system with subsystems each having respective MTTFs?
09. Please summarize the key design principles of RAID 0 through RAID 6.
10. Please list the equations for processor performance such as CPU time, CPI, IPC, and Clock Cycles. Please explain also how clock cycle time, CPI, and instruction count affect CPU time.

APPENDIX B & CHAPTER 2 - MEMORY HIERARCHY

11. What are the design principles of direct mapped, fully associative, and set associative cache organization?
12. Please provide examples to showcase address mapping policies in different types

of cache organization. You may consider a byte-addressed system for example. The examples should also specify which memory addresses to access, cache size, block size, and cache associativity, and explain how to use such factors to accurately map a memory address to a cache set or a cache line.

13. Please provide various versions of equations for computing average memory access time given a single-level cache and a multi-level cache.
14. Please provide the key ideas for various cache optimization techniques and how they affect performance such as hit rate, miss rate, hit time, access latency, miss penalty, and power consumption.
15. Please describe the process of address translation given a two-level cache with the help of TLB.

*** APPENDIX A & APPENDIX C & CHAPTER 3 - ILP***

16. What are the three types of dependences?
17. What are the common types of hazards? Please explain also the relationship between data dependence and data hazard.
18. What are commonly used forwarding solutions to deal with data hazards?
19. Please provide examples of commonly used branch predictors and analyze their properties about whether one or more branches are considered and about the storage cost.
20. Please explain the idea of superscalar and how it helps to increase IPC or decrease CPI.
21. Please describe the concrete operations (especially about when and how different types of hazards are processed) in each step for Scoreboarding, Tomasulo's Algorithm, and Tomasulo's Algorithm with speculation.
22. Please provide a sequence of instructions, then determine the timelines for different dynamic-scheduling algorithms to pipeline them. Please specify also necessary assumptions such as whether forwarding is supported, whether certain operations (such as issuing an instruction and capturing a broadcast result) can be executed in the same cycle.

CHAPTER 4 - DLP

23. What are the three types of commonly used DLP architectures?
24. What are the working principles of the vector architecture?
25. What is the key idea of dynamic register typing?
26. How to compute vector execution time? What factors does it depend on? Why does chaining helps to optimize vector execution time?
27. Please provide an example to showcase how stride affects memory access latency in a system with multi-banked memory.
28. Please provide an example program to showcase loop-carried dependences and how to rewrite this program to eliminate such dependences.

CHAPTER 5 - TLP

29. How does communication cost in a multicore system affect system performance?

30. What is the condition for a memory system to be coherent?
31. Please provide examples to showcase how a multicore system processes a sequence of memory access requests with correct coherence-state transition using the Snooping-based coherence protocol and the directory-based coherence protocol. The examples should specify system features such as core count, cache organization, and initial cache states per core.
32. Please specify the types of orderings required by different types of consistency policies.